

MammoClaw: Towards Skill Evolving Agent Harness for Breast Cancer Mammography Analysis

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Contribution

We equip a frozen LLM with (1) mammogram-specific tools for agentic investigation and (2) a skill-evolution mechanism that distils failed trajectories into reusable skills, improving the agent's reasoning in a non-parametric manner.

Agent Skill Evolution Framework

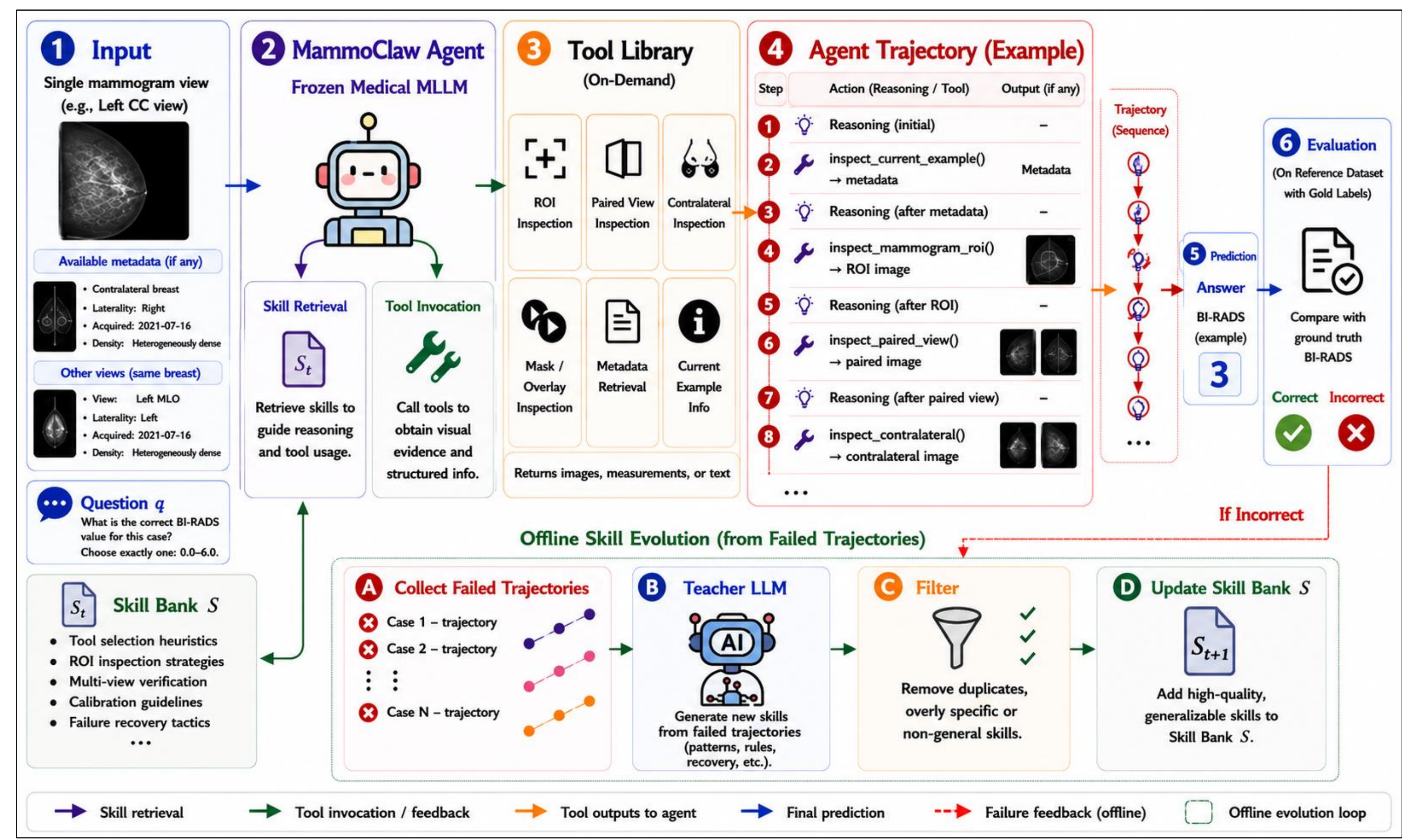


Figure 1. A frozen medical MLLM investigates a mammogram with on-demand tools and a retrievable **Skill Bank**; a teacher LLM distills failed trajectories into new skills offline.

7+ Tools to perform agentic investigation, including knowledge retrieval, ROI Tools, Multi-viewer Tools

Tool: inspect_mammogram_roi

Category: Single-image inspection

Description: Crop and enlarge a suspicious region from the current mammogram using normalized 0–1000 image-grid coordinates.

Input: bbox=[200, 200, 800, 800], description="central breast region to assess whether dense tissue appears as scattered islands (B) or confluent/heterogeneous regions (C)", enhance_contrast=true

Output: ROI crop attached (right)

Tool: inspect_paired_mammogram_view

Category: Cross-view comparison

Description: Inspect the alternate projection of the same breast by returning paired CC and MLO views side by side.

Input: (no arguments)

Output: side-by-side panel, current L CC (left) vs. paired L MLO (right)

Tool: inspect_contralateral_breast

Category: Cross-breast comparison

Description: Inspect the current breast with the contralateral breast from the same subject, source dataset, and view projection.

Input: (no arguments)

Output: side-by-side panel, current L MLO (left) vs. contralateral R MLO (right), for global asymmetry comparison.

Figure 2. Example tools from the library: **ROI** crop, **paired CC/MLO** view and **contralateral-breast** comparison.

Qualitative Results and Insights

Method	Tools	Skills	Macro-F1 [95% CI]
Baseline	×	×	0.108 [0.103, 0.113]
MammoClaw	✓	×	0.106 [0.093, 0.122]
MammoClaw	✓	✓	0.148 [0.121, 0.180]

Table 1. BI-RADS Macro-F1 (95% CI). Tools alone do not help; adding evolved skills lifts Macro-F1 from 0.106 to **0.148**.

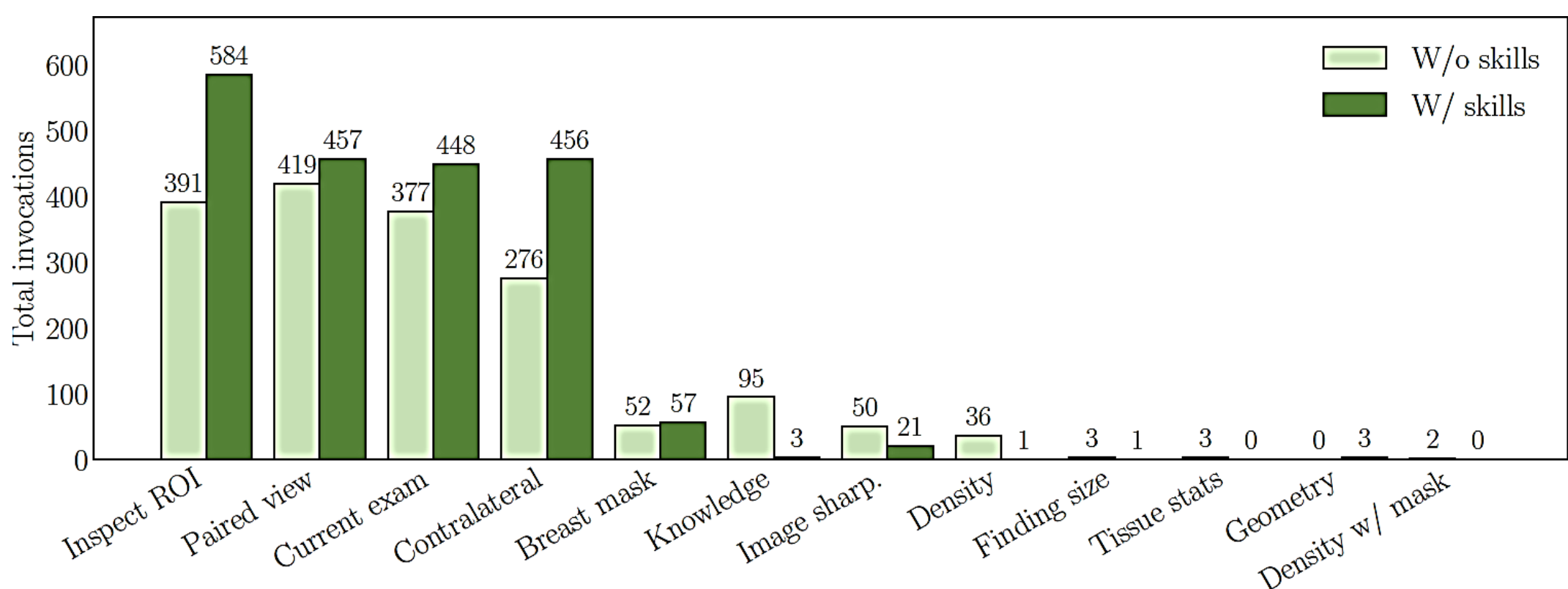


Figure 3. Tool invocations. Skills shift calls toward core inspection tools (ROI, paired view, contralateral) and away from rarely useful ones.

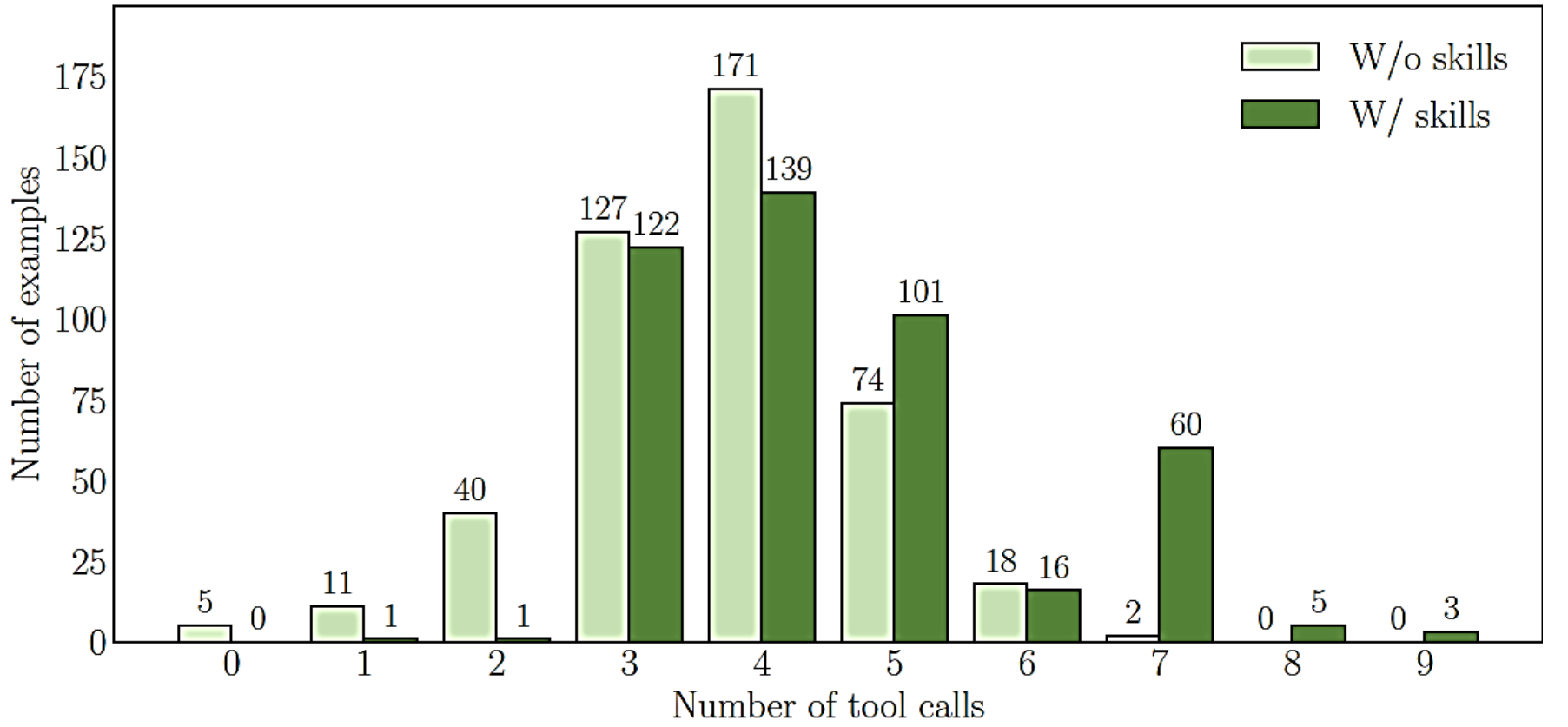


Figure 4. Tool calls per example. With skills the agent rarely stops early (0-2 calls) and investigates longer (5-7 calls).

Evolved Mammography Skills

Metadata

name: verify-suspicious-findings-with-multi-view-comparison

description: Before concluding a suspicious finding like spiculation or architectural distortion, always compare with the paired view and contralateral breast to rule out artifacts or normal tissue overlap.

metadata:

category: general

source: dynamic

task: BI-RADS

Verify Suspicious Findings with Multi-view Comparison

1. Immediately invoke `inspect_paired_mammogram_view` to see if the finding is visible in both CC and MLO projections. If it is not clearly present in both, it may be a summation artifact.
2. Then invoke `inspect_contralateral_breast` to check for bilateral symmetry. A finding that appears in the same location on the opposite breast is likely a normal anatomic variant.
3. Only after confirming the finding is unilateral and present on two orthogonal views should you assign a BI-RADS category of 4 or 5.
4. If the finding disappears or appears symmetric, reassess as benign or normal (BI-RADS 1 or 2).

Anti-pattern

- Labeling a single-view finding as suspicious without multi-view confirmation.
- Skipping contralateral comparison when suspecting pathology.

Figure 5. An example evolved skill (SKILL.md): verify suspicious findings with paired-view and contralateral comparison before assigning BI-RADS 4 or 5.

Agent Trajectory

Retrieved Skills (injected into system prompt before Step 1)

verify-suspicious-findings-with-multi-view-comparison • rule-out-artifact-with-contralateral-comparison • calibrate-bi-rads-using-feature-checklist • distinguish-benign-from-probably-benign-calcifications • systematic-quadrant-inspection-for-subtle-findings • tool-execution-before-conclusion • avoid-characterizing-findings-without-tool-confirmation • structured-reasoning-grounded-in-tool-outputs • use-uncertainty-resolution-sequence

User Query

Q: "What is the correct BI-RADS value for this case? Choose exactly one: 0.0–6.0."

Step 1 — Reasoning (before any tool call)

"The user wants me to determine the BI-RADS category for this mammogram. Let me start by examining the image and the metadata. Looking at the image, I can see a mammogram with a large, well-circumscribed, round/oval mass in the upper outer region. The mass appears to have relatively uniform density and smooth margins. This could be a benign finding like a fibroadenoma or a cyst. Let me first check the metadata to understand what view this is and get more context."

Step 1 — Tool Call: inspect_current_example()

Output: laterality=L, view=CC, age=46, paired_view=MLO, contralateral=kau-bcmd_691.jpg

Step 2 — Reasoning (after metadata, before ROI call)

"I see a mammogram of the left breast (CC view). There is a large, well-circumscribed, round/oval mass in the upper outer region of the breast. It appears to have a relatively homogeneous density, possibly slightly less dense than the surrounding fibroglandular tissue. The margins appear well-defined and circumscribed. Let me first inspect this finding more closely with an ROI, then check the paired view and contralateral breast to confirm it's a real finding and characterize it properly."

Step 2 — Tool Call: inspect_mammogram_roi(bbox=[450,100,850,550])

Tool returns: contrast-enhanced crop of the marked region (no automated interpretation — the model reads the image itself, below).

Step 3 — Reasoning (after ROI, before paired-view call)

"The ROI shows a large, well-circumscribed, oval-shaped mass. The margins appear very smooth and well-defined. The density is relatively homogeneous. This appearance is highly suggestive of a benign lesion, such as a fibroadenoma or a cyst. To confirm this and rule out any subtle spiculation or asymmetry, I need to check the paired view (MLO) to see if the mass is visible and maintains its benign features, and check the contralateral breast to ensure there's no similar finding. Let's start with the paired view."

Step 3 — Tool Call: inspect_paired_mammogram_view()

Tool returns: side-by-side panel — current L CC (left) vs. paired L MLO (right). The model sees this joint image and reasons over it below.

Step 4 — Reasoning (after paired view, before contralateral call)

"The paired view shows the mass is present in both the CC and MLO projections. It appears as a large, well-circumscribed, round/oval mass. The margins look very smooth and well-defined. This is a classic appearance of a benign mass, likely a fibroadenoma or a cyst. Let me check the contralateral breast to see if there's a similar finding, although a unilateral well-circumscribed mass is still likely benign. Let me check the contralateral breast to be thorough, as per the guidelines."

Step 4 — Tool Call: inspect_contralateral_breast()

Tool returns: side-by-side panel — current L CC (left) vs. contralateral R CC (right); the model reads this joint image to check for a mirrored finding.

Step 5 — Final Decision ✓ (matches ground truth BI-RADS 3)

"The mass is visible in both the CC and MLO projections, confirming it is a real 3D structure and not a summation artifact. It is unilateral, as no corresponding mass is seen in the contralateral right breast. The mass has well-defined margins and a homogeneous soft-tissue density, lacking spiculations, architectural distortion, or suspicious calcifications. It does not meet the criteria for BI-RADS 2 (which requires classic benign features like fat density) nor BI-RADS 4/5 (which require suspicious features)." ⇒ BI-RADS 3 (probably benign; short-interval follow-up).

Figure 6. Retrieved skills are injected in to the system prompt; the agent then inspects metadata, the suspected ROI, the paired view and the contralateral breast before committing to BI-RADS 3, matching the ground-truth.

Key Takeaways

- Tools alone do not help: Macro-F1 0.106 vs. 0.108 for the baseline.
- Evolved skills lift it to 0.148 [0.121, 0.180] F1.
- Skills change behavior: more investigations; deeper reflection; leverage multi-views

